

Total Effective Carbon Price

A CES-Based Metric for Comparing Climate Policy Instruments

Working Paper — Draft

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1. Executive Summary

Every climate policy changes fuel prices. Some policies make dirty fuels more expensive relative to clean ones (**fuel switching**). Some raise the overall cost of energy (**demand reduction**). Most do both. These effects vary across sectors and fuels.

The **total effective carbon price** (TECP), denoted τ^* , answers one question: *what single economy-wide carbon tax would reduce emissions by the same amount as the actual policy mix?*

$$\tau^* = \frac{1}{\Lambda} \sum_s \sum_f E_{fs} \left[\underbrace{\tilde{\varepsilon}_{fs}^{\text{int}} \cdot \tau_{fs}^{\text{int}}}_{\text{switching}} + \underbrace{\tilde{\varepsilon}_s^{\text{ext}} \cdot \tau_s^{\text{ext}}}_{\text{demand}} \right]$$

The tilde $\tilde{\varepsilon}$ denotes **semi-elasticities**: the proportional change in emissions per *unit* change in carbon price (\$/tCO₂), not per percentage change. Semi-elasticities are the natural form when

the policy variable is an absolute price.

s, f	Sector and fuel indices.
E_{fs}	Baseline emissions from fuel f in sector s (tCO ₂).
$\tilde{\varepsilon}_{fs}^{\text{int}}$	Intensive semi-elasticity: proportional shift in fuel f 's consumption per \$/tCO ₂ of relative price signal. Fuel- and sector-specific.
τ_{fs}^{int}	Intensive carbon price on fuel f in sector s : the component of the policy that changes this fuel's price <i>relative to others</i> , in \$/tCO ₂ .
$\tilde{\varepsilon}_s^{\text{ext}}$	Extensive semi-elasticity: proportional reduction in total energy demand per \$/tCO ₂ of average cost increase. Sector-level.
τ_s^{ext}	Extensive carbon price in sector s : the component that raises the <i>average</i> cost of energy, in \$/tCO ₂ .
Λ	Normalisation constant: emissions reduction from a \$1/tCO ₂ uniform carbon tax. Depends only on the energy balance, prices, and elasticities—not on the policy. Computed once per country.

Intuition. The numerator sums the emissions reduction from every fuel in every sector, split into two channels. Each channel is a semi-elasticity times a carbon price. The denominator Λ converts the total reduction into “equivalent dollars of carbon tax.”

Key property. If the policy *is* a uniform carbon tax at rate τ , then $\tau_{fs}^{\text{int}} = \tau_s^{\text{ext}} = \tau$ for all f, s , and $\tau^* = \tau$ exactly.

Examples. A revenue-neutral feebate has $\tau_s^{\text{ext}} \approx 0$: only the switching channel contributes. A uniform energy tax has $\tau_{fs}^{\text{int}} = 0$: only the demand channel contributes. The TECP reveals how much abatement each instrument delivers per dollar, and allows heterogeneous policy packages to be compared on a common scale.

2. Motivation and Scope

Countries use very different combinations of climate policy instruments: carbon taxes, emissions trading systems, feebates, fuel excise duties, clean fuel standards, and subsidies. Comparing the stringency of these policy packages requires a common metric. Existing metrics—such as the OECD's Effective Carbon Rates or the IMF's carbon pricing scores—typically aggregate the explicit and implicit carbon price signals but do not weight them by their effectiveness in reducing emissions.

The TECP addresses this gap. It weights each policy instrument by its *marginal abatement effectiveness*, accounting for the fact that a dollar of fuel excise does not reduce emissions by the same amount as a dollar of carbon tax. The weighting is derived from the structure of the energy system—fuel mix, prices, emission factors, and substitution possibilities—using a CES (constant elasticity of substitution) demand model.

The TECP is a *static, long-run equilibrium* metric. It does not incorporate adjustment dynamics or time horizons. This is a deliberate design choice: the purpose is cross-country comparability, not forecasting. Adjustment dynamics are handled separately in a companion partial adjustment model (see Paper II).

3. Notation and Units

Symbol	Definition	Units
f	Fuel index	—
s	Sector index	—
q_{fs}	Baseline energy consumption of fuel f in sector s	GJ
p_{fs}	Pre-tax price of fuel f in sector s	\$/GJ
e_f	Emission factor of fuel f	tCO ₂ /GJ
ϕ_{fs}	Technical efficiency of fuel f in sector s	—
$\hat{p}_{fs} = p_{fs}/\phi_{fs}$	Effective price (cost per unit of service)	\$/GJ _{svc}
$\hat{e}_{fs} = e_f/\phi_{fs}$	Effective emission factor (emissions per unit of service)	tCO ₂ /GJ _{svc}
$E_{fs} = q_{fs} e_f$	Baseline emissions from fuel f in sector s	tCO ₂
$\hat{w}_{fs}$	Service share of fuel f in sector s	—
$\bar{e}_s = \sum_f \hat{w}_{fs} \hat{e}_{fs}$	Service-weighted mean effective emission factor	tCO ₂ /GJ _{svc}
σ_s	CES substitution elasticity in sector s	—
η_s	Service demand elasticity in sector s	—
C_s	CES unit cost index in sector s	\$/GJ _{svc}
τ_{fs}^{int}	Intensive carbon price on fuel f in sector s	\$/tCO ₂
τ_s^{ext}	Extensive carbon price in sector s	\$/tCO ₂
τ^*	Total effective carbon price	\$/tCO ₂

4. Technical Efficiency

Different fuels deliver different amounts of useful energy service per GJ of input energy. An electric motor converts approximately 85% of input energy to useful work, while an internal combustion engine converts approximately 25%. A GJ of electricity therefore delivers roughly three times the transport service of a GJ of gasoline.

The CES demand model should operate over **effective** (service-equivalent) units, not raw energy. Each fuel carries a technical efficiency parameter ϕ_{fs} that converts physical energy to service-equivalent energy:

$$\text{Effective quantity: } \hat{q}_{fs} = \phi_{fs} \cdot q_{fs} \quad (1)$$

$$\text{Effective price: } \hat{p}_{fs} = p_{fs}/\phi_{fs} \quad (2)$$

$$\text{Effective emission factor: } \hat{e}_{fs} = e_f/\phi_{fs} \quad (3)$$

Service shares, the CES cost index, and all semi-elasticities are computed over effective quantities and prices. This ensures that the model correctly captures the competitive position of high-efficiency fuels (such as electricity in transport) that appear expensive per GJ but are cheap per unit of service delivered.

Throughout the remainder of this paper, all prices, emission factors, and shares should be understood as effective (service-equivalent) quantities unless otherwise noted. The hat notation is dropped for readability.

5. Semi-Elasticities from the CES

5.1 Isoelastic elasticities to semi-elasticities

An isoelastic elasticity gives the proportional response to a proportional price change: $\varepsilon = \partial \ln q / \partial \ln p$. A semi-elasticity gives the proportional response to an *absolute* price change: $\tilde{\varepsilon} = \partial \ln q / \partial p$. Since $\partial \ln p = \partial p / p$:

$$\tilde{\varepsilon} = \frac{\varepsilon}{p} \quad (4)$$

5.2 CES parameters to semi-elasticities

From a CES with substitution elasticity σ_s and service demand elasticity η_s :

Intensive semi-elasticity (fuel f , sector s):

$$\tilde{\varepsilon}_{fs}^{\text{int}} = \sigma_s (1 - \hat{w}_{fs}) \frac{\hat{e}_{fs} - \bar{\hat{e}}_s}{\hat{p}_{fs}} \quad (5)$$

This combines four factors: σ_s (how substitutable fuels are), $(1 - \hat{w}_{fs})$ (room to substitute—a dominant fuel has less scope), $(\hat{e}_{fs} - \bar{\hat{e}}_s)$ (carbon intensity relative to the sector average, in service-equivalent terms), and $1/\hat{p}_{fs}$ (conversion from elasticity to semi-elasticity).

Extensive semi-elasticity (sector s):

$$\tilde{\varepsilon}_s^{\text{ext}} = |\eta_s| \frac{\bar{\hat{e}}_s}{C_s} \quad (6)$$

This combines $|\eta_s|$ (demand responsiveness), $\bar{\hat{e}}_s$ (average carbon intensity of service), and $1/C_s$ (conversion from elasticity to semi-elasticity).

6. Policy Price Decomposition

Any policy that changes fuel prices by Δp_{fs} (in \$/GJ) can be decomposed into extensive and intensive components. Working in effective (service-equivalent) terms, define $\Delta \hat{p}_{fs} = \Delta p_{fs} / \phi_{fs}$.

Extensive component (average cost increase in sector s):

$$\tau_s^{\text{ext}} = \frac{\sum_f \hat{w}_{fs} \Delta \hat{p}_{fs}}{\bar{\hat{e}}_s} \quad (7)$$

This converts the service-share-weighted average effective price increase into carbon-price-equivalent units by dividing by the mean effective emission factor.

Intensive component (relative price signal for fuel f):

$$\tau_{fs}^{\text{int}} = \frac{\Delta \hat{p}_{fs} - \sum_{f'} \hat{w}_{f's} \Delta \hat{p}_{f's}}{\hat{e}_{fs} - \bar{\hat{e}}_s} \quad (8)$$

This is the deviation of fuel f 's effective price change from the sectoral average, normalised by the deviation of its effective emission factor from the sectoral mean.

For a pure carbon tax at rate τ , $\Delta p_{fs} = \tau \cdot e_f$, giving $\Delta \hat{p}_{fs} = \tau \cdot e_f / \phi_{fs} = \tau \cdot \hat{e}_{fs}$. Substituting confirms $\tau_{fs}^{\text{int}} = \tau_s^{\text{ext}} = \tau$ for all f, s .

7. Emissions Reduction and the TECP Formula

7.1 Emissions reduction by fuel and sector

The linearised emissions reduction from fuel f in sector s :

$$\Delta E_{fs} = E_{fs} \left[\underbrace{\tilde{\varepsilon}_{fs}^{\text{int}} \cdot \tau_{fs}^{\text{int}}}_{\text{intensive margin}} + \underbrace{\tilde{\varepsilon}_s^{\text{ext}} \cdot \tau_s^{\text{ext}}}_{\text{extensive margin}} \right] \quad (9)$$

Total emissions reduction:

$$\Delta E = \sum_s \sum_f \Delta E_{fs} \quad (10)$$

7.2 Benchmark: \$1 carbon tax

Under a uniform \$1/tCO₂ carbon tax, $\tau_{fs}^{\text{int}} = \tau_s^{\text{ext}} = 1$ for all f, s :

$$\Lambda \equiv \Delta E^{\text{bench}} = \sum_s \sum_f E_{fs} \left[\tilde{\varepsilon}_{fs}^{\text{int}} + \tilde{\varepsilon}_s^{\text{ext}} \right] \quad (11)$$

7.3 The TECP

The total effective carbon price is:

$$\tau^* = \frac{\Delta E}{\Lambda} = \frac{\sum_s \sum_f E_{fs} \left[\tilde{\varepsilon}_{fs}^{\text{int}} \tau_{fs}^{\text{int}} + \tilde{\varepsilon}_s^{\text{ext}} \tau_s^{\text{ext}} \right]}{\sum_s \sum_f E_{fs} \left[\tilde{\varepsilon}_{fs}^{\text{int}} + \tilde{\varepsilon}_s^{\text{ext}} \right]} \quad (12)$$

Equivalently, defining **abatement weights**:

$$\Omega_{fs}^{\text{int}} = E_{fs} \cdot \tilde{\varepsilon}_{fs}^{\text{int}} \quad (13)$$

$$\Omega_s^{\text{ext}} = \tilde{\varepsilon}_s^{\text{ext}} \sum_f E_{fs} \quad (14)$$

the TECP becomes:

$$\tau^* = \frac{\sum_s \left[\sum_f \tau_{fs}^{\text{int}} \cdot \Omega_{fs}^{\text{int}} + \tau_s^{\text{ext}} \cdot \Omega_s^{\text{ext}} \right]}{\sum_s \left[\sum_f \Omega_{fs}^{\text{int}} + \Omega_s^{\text{ext}} \right]} \quad (15)$$

The Ω weights are *properties of the economy*, not of the policy. They are computed once from the energy balance, fuel prices, emission factors, technical efficiencies, and CES elasticities. The policy enters only through τ_{fs}^{int} and τ_s^{ext} .

7.4 Compact form

Dropping indices and writing sums implicitly:

$$\tau^* = \frac{\sum E \left[\tilde{\varepsilon}^{\text{int}} \cdot \tau^{\text{int}} + \tilde{\varepsilon}^{\text{ext}} \cdot \tau^{\text{ext}} \right]}{\sum E \left[\tilde{\varepsilon}^{\text{int}} + \tilde{\varepsilon}^{\text{ext}} \right]} \quad (16)$$

8. Special Cases

Pure carbon tax at rate τ : $\tau_{fs}^{\text{int}} = \tau_s^{\text{ext}} = \tau$ for all f, s , so $\tau^* = \tau$. The formula is self-consistent.

Revenue-neutral feebate: $\tau_s^{\text{ext}} \approx 0$ (the average price increase is near zero), so only the intensive terms survive. The TECP equals the feebate rate scaled by the ratio of intensive-margin effectiveness to total carbon-tax effectiveness.

Uniform energy tax (same \$/GJ on all fuels): $\tau_{fs}^{\text{int}} = 0$ (no relative price signal), so only the extensive terms survive. The TECP reflects only the demand reduction channel.

Single sector, single fuel: σ is irrelevant (no switching possible). τ^* reduces to the extensive component only, equalling the fuel price increase divided by the emission factor.

Fuel subsidy: A subsidy is a negative price change. A fuel subsidy on a high-carbon fuel generates a *negative* TECP contribution, correctly reflecting that it works against emissions reduction.

9. Decomposition and Interpretation

The TECP can be decomposed along two dimensions:

By margin: The intensive contribution τ_{int}^* and extensive contribution τ_{ext}^* sum to τ^* :

$$\tau_{\text{int}}^* = \frac{\sum_s \sum_f \Omega_{fs}^{\text{int}} \tau_{fs}^{\text{int}}}{\sum_s \left[\sum_f \Omega_{fs}^{\text{int}} + \Omega_s^{\text{ext}} \right]} \quad (17)$$

$$\tau_{\text{ext}}^* = \frac{\sum_s \Omega_s^{\text{ext}} \tau_s^{\text{ext}}}{\sum_s \left[\sum_f \Omega_{fs}^{\text{int}} + \Omega_s^{\text{ext}} \right]} \quad (18)$$

By sector: The contribution of sector s to the overall TECP is:

$$\tau_s^* = \frac{\sum_f \Omega_{fs}^{\text{int}} \tau_{fs}^{\text{int}} + \Omega_s^{\text{ext}} \tau_s^{\text{ext}}}{\Lambda} \quad (19)$$

These decompositions reveal *why* two countries with the same TECP might achieve it through very different mechanisms, and which sectors or margins offer the most abatement potential per dollar of policy effort.

10. Inputs and Computation

Given (from energy balance and price data):

- q_{fs} — energy consumption (GJ) for each fuel f in sector s
- p_{fs} — pre-tax fuel price (\$/GJ)
- e_f — emission factor (tCO₂/GJ)
- ϕ_{fs} — technical efficiency (fraction of input energy converted to useful service)

Given (from CES calibration):

- σ_s — substitution elasticity per sector
- η_s — demand elasticity per sector

Given (from policy description):

- Δp_{fs} — price change on each fuel from the policy (\$/GJ)

Computation steps:

1. Compute effective prices $\hat{p}_{fs} = p_{fs}/\phi_{fs}$ and effective emission factors $\hat{e}_{fs} = e_f/\phi_{fs}$
2. Compute service shares $\hat{w}_{fs} = q_{fs}\phi_{fs}/\sum_{f'} q_{f's}\phi_{f's}$
3. Compute $\bar{\hat{e}}_s = \sum_f \hat{w}_{fs} \hat{e}_{fs}$ and $E_{fs} = q_{fs} e_f$
4. Compute C_s from the CES cost function over effective prices
5. Compute semi-elasticities $\tilde{\epsilon}_{fs}^{\text{int}}$ and $\tilde{\epsilon}_s^{\text{ext}}$
6. Decompose $\Delta \hat{p}_{fs} = \Delta p_{fs}/\phi_{fs}$ into τ_{fs}^{int} and τ_s^{ext}
7. Compute Ω weights, Λ , and τ^*

Output: τ^* (\$/tCO₂) — the total effective carbon price.

11. Relationship to the Time-Dynamic Model

The TECP is a static metric: it uses long-run equilibrium elasticities and does not incorporate adjustment dynamics. This is appropriate for cross-country comparison.

For year-by-year emissions projections, the companion partial adjustment model (Paper II) applies the same CES structure but introduces an adjustment speed parameter λ that governs how quickly actual quantities converge to equilibrium. The TECP uses the long-run elasticities (σ, η) directly; the partial adjustment model effectively uses short-run elasticities ($\lambda\sigma, \lambda\eta$) in any given year.

The TECP can therefore be interpreted as the carbon price equivalence that would hold once the economy has fully adjusted. The partial adjustment model shows the path to that equilibrium.